

Name: _____

Date: _____

Standards: MA.5.FR.1.1; MA.5.FR.2.1-2.4; MA.5.AR.1.2-1.3

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 04: addition and subtraction of fractions.

1 Fractions and Division

Write each division as a fraction or mixed number.

$$11 \div 4 = \underline{\hspace{2cm}} \quad | \quad 7 \div 3 = \underline{\hspace{2cm}} \quad | \quad 9 \div 6 = \underline{\hspace{2cm}}$$

Answers _____

2 Add Unlike Fractions

Add. Simplify.

$$\frac{2}{3} + \frac{1}{4} = \underline{\hspace{2cm}} \quad | \quad \frac{5}{6} + \frac{3}{8} = \underline{\hspace{2cm}} \quad | \quad \frac{3}{5} + \frac{7}{10} = \underline{\hspace{2cm}}$$

Answers _____

3 Subtract Unlike Fractions

Subtract. Simplify.

$$\frac{7}{8} - \frac{1}{3} = \underline{\hspace{2cm}} \quad | \quad \frac{5}{6} - \frac{3}{4} = \underline{\hspace{2cm}} \quad | \quad \frac{4}{5} - \frac{7}{15} = \underline{\hspace{2cm}}$$

Answers _____

7 Find the Mistake

Leo subtracts $6 \frac{1}{3} - 2 \frac{3}{4}$ and writes: LCD = 12; $6 \frac{4}{12} - 2 \frac{9}{12}$. He then regroupes $6 \frac{4}{12}$ as $5 \frac{12}{12} + \frac{4}{12} = 5 \frac{16}{12}$, and subtracts to get $3 \frac{7}{12}$.

Is Leo's answer correct? Check every step and identify any error.

Verification and Correction _____

8 Subtract from a Whole Number

Subtract.

$$9 - 4 \frac{5}{8} = \underline{\hspace{2cm}} \quad | \quad 12 - 7 \frac{2}{3} = \underline{\hspace{2cm}} \quad | \quad 6 - 1 \frac{7}{12} = \underline{\hspace{2cm}}$$

Answers _____

9 Three-Step Problem

Kai had 10 feet of wire. He cut off $3 \frac{3}{4}$ feet for one project and $2 \frac{5}{6}$ feet for another.

How much wire is left? Write an expression and solve step by step.

Expression and Answer _____

10 Prove It

Atlas says: 'The sum of any two unlike fractions whose numerators are both 1 will always be less than 1 if both denominators are greater than 2.'

Test Atlas's claim with $\frac{1}{3} + \frac{1}{4}$ and $\frac{1}{5} + \frac{1}{6}$. Is it always true? Can you find a counterexample?

Investigation and Explanation _____